Name- Bhumika Shrotriya

Course Name – Data Analytics With GenAI

Subject – Data Processing & ETL

Topic – Data Transformation

Question 1: Define Data Transformation in ETL and explain why it is important.

Answer 1:

Data Transformation is the second step in the ETL (Extract, Transform, Load) process where raw extracted data is converted into a suitable format for the target system. This involves cleaning, restructuring, aggregating, and enriching the data before loading it into a data warehouse or database.

It is important because raw data from multiple sources is often inconsistent, incomplete, or in different formats. Transformation ensures data quality, consistency, and compatibility, making it reliable for analysis and decision-making.

Question 2: List any four common activities involved in Data Cleaning.

Answer 2:

1. Handling missing values (imputation or removal)
2. Removing duplicate records
3. Correcting inconsistent formats (e.g., date formats, gender labels)
4. Detecting and treating outliers

Question 3: What is the difference between Normalization and Standardization?

Answer 3:

Normalization (Min-Max Scaling) rescales data to a fixed range, typically [0, 1], using the formula: X_new = (X - X_min) / (X_max - X_min). It is best used when the data does not follow a normal distribution and when you need bounded values.

Standardization (Z-score Scaling) transforms data to have a mean of 0 and a standard deviation of 1, using: X_new = (X - mean) / std. It is preferred when the data follows a roughly normal distribution or when outliers are present, as it is less sensitive to extreme values.

Question 4: Two techniques to handle missing values in the "Age" column.

Answer 4:

Technique 1 — Mean/Median Imputation: Replace missing values with the mean (if data is normally distributed) or median (if data is skewed). Use this when the proportion of missing data is small and the missing values are random.

Technique 2 — Deletion (Row Removal): Remove rows where "Age" is missing. Use this only when the number of missing records is very small (e.g., less than 5%) and removing them won't introduce bias or significantly reduce the dataset size.

Question 5: Standardized Gender entries.

Answer 5:

Original	Standardized
"M"	"Male"
"male"	"Male"
"F"	"Female"
"Female"	"Female"
"MALE"	"Male"
"f"	"Female"

The logic: map all variations of "M", "male", "MALE" → "Male" and all variations of "F", "f", "Female" → "Female", typically using lowercasing and a lookup/mapping dictionary.

Question 6: What is One-Hot Encoding? Example with Red, Blue, Green.

Answer 6:

One-Hot Encoding is a technique to convert categorical variables into a binary (0/1) numerical format. Each unique category becomes a new column, and a 1 is placed in the column corresponding to the category present, with 0s in all others.

Example with categories Red, Blue, Green:

Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

This avoids implying any ordinal relationship between categories, which would happen if they were simply label-encoded as 1, 2, 3.

Question 7: Difference between Data Integration and Data Mapping in ETL.

Answer 7:

Data Integration is the broader process of combining data from multiple, heterogeneous sources (databases, APIs, files, etc.) into a unified, consistent view. It focuses on merging and consolidating data from different systems.

Data Mapping is a specific step within integration that defines how fields/columns from the source system correspond to fields in the target system. It is essentially the blueprint or rules that guide how source data elements are translated, transformed, and linked to destination data elements.

In short: Data Integration is the "what and why" (combining sources), while Data Mapping is the "how" (field-to-field correspondence rules).

Question 8: Why Z-score Standardization is preferred over Min-Max Scaling when outliers exist.

Answer 8:

Min-Max Scaling is heavily influenced by the minimum and maximum values in the dataset. When outliers are present, they become the min or max, which compresses all other values into a very narrow range, distorting the distribution and reducing the usefulness of the scaled data.

Z-score Standardization, on the other hand, uses the mean and standard deviation. Outliers do affect the mean and std slightly, but the impact is spread across all values rather than compressing the entire range. Most normal data points still get meaningful, well-spread scaled values. Additionally, Z-score doesn't restrict data to a fixed range, so outliers don't dominate the scaling. This makes it a more robust choice when the dataset contains extreme values.